

# TUTORIUM 8.

PS.1:  $n \in \mathbb{R}^3$ ,  $\|n\|=1$ ,  $\mathbb{R}^{3 \times 3} \ni A: Ax = n \times x$ ,  $\forall x \in \mathbb{R}^3$

① ZZ:  $A = \begin{pmatrix} 0 & -n_3 & n_2 \\ n_3 & 0 & -n_1 \\ -n_2 & n_1 & 0 \end{pmatrix}$   
Bew:

$$Ax = n \times x = \begin{pmatrix} n_2 x_3 - n_3 x_2 \\ n_3 x_1 - n_1 x_3 \\ n_1 x_2 - n_2 x_1 \end{pmatrix} = \begin{pmatrix} 0 & -n_3 & n_2 \\ n_3 & 0 & -n_1 \\ -n_2 & n_1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

② ZZ.  $e^{tA} = (n \cdot x)n + (x - (n \cdot x)n) \cos t + (n \times x) \sin t$

Bew:  $t \mapsto e^{tA}$  eindeutige Lösung  $\dot{x} = Ax$ ,  $x(0) = x_0$   
 $\dot{x} = n \times x$ ,  $e^{0A} = \mathbb{1}$

$$x(t) = (n \cdot x)n + (x - (n \cdot x)n) \cos t + (n \times x) \sin t$$

$$e^{tA} = \sum_{k=0}^{\infty} \frac{1}{k!} (tA)^k = \mathbb{1} + tA + \frac{t^2}{2!} A^2 + \frac{t^3}{3!} A^3 + \dots$$

$$a \times (b \times c) = b(a \cdot c) - c(a \cdot b)$$

$$Ax = n \times x$$

$$A^2 x = A(Ax) = A(n \times x) = n \times (n \times x) = n(n \cdot x) - x(n \cdot n) = \underbrace{(n \cdot x)n}_{x_{||}} - x = -x_{\perp}$$

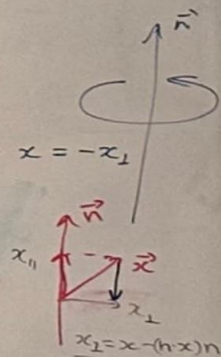
$$A^3 x = A(A^2 x) = n \times [(n \cdot x)n - x] = 0 - n \times x$$

$$A^4 x = A(A^3 x) = -n \times (n \times x) = x_{\perp}$$

$$A^5 x = A(A^4 x) = n \times x$$

$$A^6 x = -x_{\perp}$$

$\vdots$

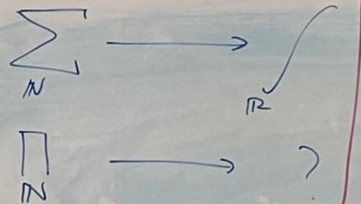


$$e^{tA} x = \left( \mathbb{1} + \frac{t^2}{2!} A^2 + \frac{t^4}{4!} A^4 + \dots \right) x + \left( tA + \frac{t^3}{3!} A^3 + \frac{t^5}{5!} A^5 + \dots \right) x$$

$$= \underbrace{(n \cdot x)n - (n \cdot x)n}_{x_{||}} + \frac{t^2}{2!} (-x_{\perp}) + \frac{t^4}{4!} x_{\perp} + \dots + \left( t(n \times x) + \frac{t^3}{3!} (-n \times x) + \frac{t^5}{5!} (n \times x) + \dots \right)$$

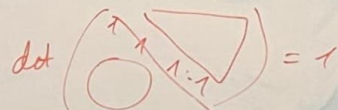
$$= (n \cdot x)n + x_{\perp} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} t^{2k} + (n \times x) \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!} t^{2m+1}$$

$$= (n \cdot x)n + x_{\perp} \cos t + (n \times x) \sin t$$



(b) Geg:  $A = \begin{pmatrix} \lambda & & & 0 \\ & \lambda & & \\ & & \ddots & \\ 0 & & & \lambda \end{pmatrix} = \lambda \mathbb{1} + N$

$e^{tN} = 1 + tN + \dots + \frac{t^N}{N!} N^N$  1 auf Diag



$\det(e^A) = \det(e^{\lambda \mathbb{1} + N}) = \det(e^{\lambda \mathbb{1}}) \cdot 1 = e^{t \operatorname{tr}(\lambda \mathbb{1})} = e^{t \operatorname{tr}(\lambda \mathbb{1} + N)} = e^{t \operatorname{tr}(A)}$

$e^{\lambda \mathbb{1} + N} = e^{\lambda \mathbb{1}} \cdot e^N$

$e^{t \operatorname{tr}(A)} = \det(e^{tA})$   
 $\lg(ab) = \lg a + \lg b$

$(a \times b)_i = \varepsilon^{ijk} a_j b_k$   
 $\begin{cases} 1 & , ijk \text{ gerade } 123 \\ 0 & \text{keine} \\ -1 & \text{ungerade } 123 \end{cases}$

P8.2:  $A \in \mathbb{R}^{n \times n}$

z.z.:  $\det(e^{tA}) = e^{t \operatorname{tr}(A)}$

Geg.:  $A$  diag.  $\exists S \in GL(n)$   $\wedge D = \operatorname{diag}(\lambda_1, \dots, \lambda_n) \Rightarrow A = SDS^{-1} \Rightarrow S^{-1}AS = D$

$\det(S) \det(S^{-1}) = \det(SS^{-1}) = 1$

$\det(e^{tA}) = \det(S^{-1} e^{tA} S) = \det(e^{t(S^{-1}AS)}) = \det(e^{tD}) = \prod_{i=1}^n e^{t \lambda_i}$   
 $= e^{t \sum_{i=1}^n \lambda_i} = \exp(t \cdot \operatorname{tr}(D)) = \exp(t \cdot \operatorname{tr}(S^{-1}AS)) = \exp(t \cdot \operatorname{tr}(ASS^{-1})) = \exp(t \cdot \operatorname{tr}(A))$

(c) Geg:  $A = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}$   
 $\operatorname{tr}(A) = \operatorname{tr}(A_1) + \operatorname{tr}(A_2)$   
 $\det(e^{tA}) = \det(e^{tA_1}) \cdot \det(e^{tA_2})$   
 $= e^{t \operatorname{tr}(A_1)} \cdot e^{t \operatorname{tr}(A_2)} = e^{t(\operatorname{tr}(A_1) + \operatorname{tr}(A_2))} = e^{t \operatorname{tr}(A)}$

$e^{tS^{-1}AS} = \sum_{k=0}^{\infty} \frac{t^k}{k!} (S^{-1}AS)^k = S^{-1} \sum_{k=0}^{\infty} \frac{t^k}{k!} A^k S = S^{-1} e^{tA} S$

2:  $S^{-1}AS \cdot S^{-1}AS$

3:  $(S^{-1}AS)(S^{-1}AS)(S^{-1}AS)$

$e^a \cdot e^b = e^{a+b}$

(d)  $A$  ähnlich zu  $J$ , Jordanblöcke  
 $A = M J M^{-1}$ , (e)  $\wedge (c) \det(e^{tA}) = e^{t \operatorname{tr} A}$

$\det(e^{tA}) = \det(e^{t M J M^{-1}}) = \det(M e^{tJ} M^{-1})$   
 $= \det(M) \cdot \det(e^{tJ}) \cdot \det(M^{-1}) = \det(e^{tJ}) = e^{t \operatorname{tr} J}$   
 $= e^{t \operatorname{tr}(J M^{-1} M)} = e^{t \operatorname{tr}(M J M^{-1})} = e^{t \operatorname{tr} A}$

$\ln \left( \prod_i c_i \right) = \sum_i \ln(c_i) = \int \ln(c_i) = \exp \left( \int \ln(c_i) \right) \prod_x f(x) dx$

# TUTORIUM 8.

P8.3:  $X: \mathbb{R}^d \setminus \{0\} \rightarrow \mathbb{R}^d$ ,  $X(x) = \frac{x}{\|x\|^\alpha}$ .

$$r := \|x\|_2 \quad \|x\|^\alpha \rightarrow r^\alpha \Rightarrow X(x) = \frac{x}{r^\alpha}$$

$$= \frac{x}{\sqrt{x_1^2 + \dots + x_d^2}^\alpha}$$

$$\left( \sqrt{x_1^2 + \dots + x_d^2} \right)' = \frac{1}{2 \sqrt{x_1^2 + \dots + x_d^2}} \cdot 2x_i$$

$$\operatorname{div} X = \nabla \cdot X = \sum_{i=1}^d \frac{\partial}{\partial x_i} \left( \frac{x_i}{r^\alpha} \right)$$

$$\rightarrow \frac{\partial}{\partial x_i} \left( \frac{x_i}{r^\alpha} \right) = \frac{1}{r^\alpha} + x_i \cdot \frac{\partial}{\partial x_i} \left( \frac{1}{r^\alpha} \right) = \frac{1}{r^\alpha} + x_i (-\alpha) \cdot \frac{1}{r^{\alpha+1}} \cdot \frac{2x_i}{2r} = \frac{1}{r^\alpha} - \frac{x_i^2 \alpha}{r^{\alpha+2}}$$

$$\nabla \cdot X = \sum_{i=1}^d \left( \frac{1}{r^\alpha} - \frac{x_i^2 \alpha}{r^{\alpha+2}} \right) = \frac{1}{r^\alpha} \sum_{i=1}^d 1 - \frac{\alpha}{r^{\alpha+2}} \sum_{i=1}^d x_i^2$$

$$= \frac{d}{r^\alpha} - \frac{\alpha}{r^{\alpha+2}} \cdot (x_1^2 + x_2^2 + \dots + x_d^2) = \frac{d}{r^\alpha} - \frac{\alpha}{r^{\alpha+2}} \cdot r^2$$

$$= \frac{d-\alpha}{r^\alpha} \stackrel{!}{=} 0 \Rightarrow d-\alpha=0 \Rightarrow \underline{d=\alpha}$$